

1 TRAINING

Training stays
outer fit 60%

Fit common rule
covariates + ICU time

Population quantile
common rule

2 TUNING

Tuning stays
outer fit 20%

Profiles
level, slope, summary

Tune and freeze
basis, λ , calibration

Frozen rule

3 HELD-OUT ASSESSMENT

Held-out stay
20% outer fold

Index window
early MAP

Stay-specific profile
 $\hat{u}_i(r, P_r)$

Later quantile forecast
 $\hat{q}_{\tau_i}(t)$

Later window
reserved MAP

Stay-weighted check loss
 ℓ_i

Fold mean
stay loss